

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

(192511402)

I Prithvishwaran J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Question 1

Case Study - Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. for-all x: Vehicle(x) AND EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]
- II. for-all x: ClearPath(x) $\rightarrow$ GreenSignal(x) AND NOT RedSignal(x) [Rule 2]
- III. for-all x for-all y: Vehicle(x) AND Behind(x,y) AND GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
 - 2) P2: IrrigationNeeded AND CropWheat $\rightarrow$ ApplyDripMethod
 - 3) P3: NOT ApplyDripMethod AND CropWheat $\rightarrow$ CropAtRisk
 - 4) Facts: SoilDry = True, CropWheat = True
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- 4) NOT Wumpus[x,y] AND NOT Pit[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	24
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	88
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	8

92

HP

Forward chaining:

Forward chaining starts with the known facts and applies rules whose conditions are satisfied.

Step 1:

Known facts:

F, C, B

Rule 1:

$F \wedge C \rightarrow Flu$

Patient A may have Flu:

Step 2:

Rule 2;

$F \wedge R \rightarrow M$

~~Have fever~~ but Rash is not

Step 3 :

Rule 3:

$C \wedge B \rightarrow P$

Patient A may have Pneumonia

* Possible Flu

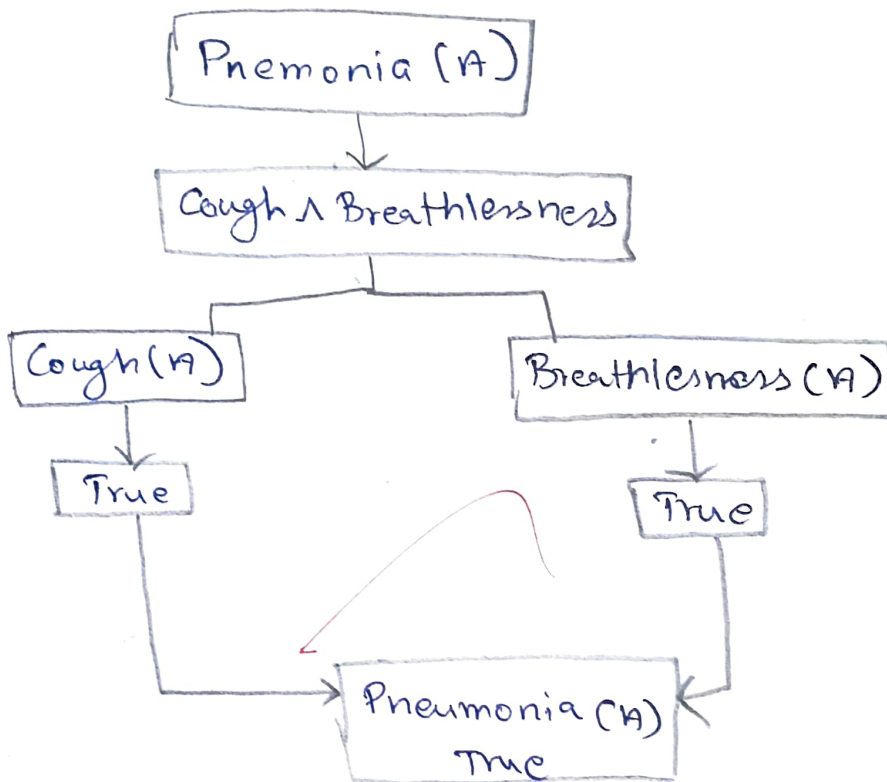
* Possible Pneumonia.

Backward chaining for Pneumonia:

Backward chaining starts with the goal and searches for a rule that can produce it

$\text{Cough}(A) \wedge \text{Breathlessness}(A) \rightarrow \text{Pneumonia}(A)$

Goal Reduction Tree



∴ Patient A has
Possible Pneumonia ∴

Autonomous Traffic Management Agent:

Unification of Rule 1 with Given facts

Rule 1:

$\forall x \text{ vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{clearPath}(x)$

Given facts:

$\text{vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Unification:

Expression

$\text{vehicle}(x)$ with
 $\text{vehicle}(\text{Ambulance})$

Substitution

$x / \text{Ambulance}$

$\text{EmergencyType}(x)$ with
 $\text{EmergencyType}(\text{Ambulance})$

$x / \text{Ambulance}$

$\text{clearPath}(x)$

$\text{clearPath}(\text{Ambulance})$

MOU:

$\theta = \{x / \text{Ambulance}\}$

After applying substitution

$\text{clearPath}(\text{Ambulance})$

Resolution Proof of Proceed (CarA)

Relevant Rules

Rule 1:

$\text{vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{clearpath}(x)$

Rule 2:

$\text{clearpath}(x) \rightarrow \text{Green Signal}(x) \wedge \neg \text{Red Signal}(x)$

Rule 3:

$\text{vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{Green Signal}(y) \rightarrow \text{Proceed}(x)$

Given facts:

$\text{vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{vehicle}(\text{Car A})$

$\text{Behind}(\text{Car A}, \text{Ambulance})$

Step 1: Convert Rules to CNF:

Rule 1

$$A \rightarrow B \equiv \neg A \vee B$$

Rule 2:

- $\rightarrow \text{Clear path}(x) \vee \text{Green Signal}(x)$
- $\rightarrow \text{Clear path}(x) \vee \rightarrow \text{Red Signal}(x)$

Rule 3:

CNF:

- $\rightarrow \text{Vehicle}(x) \vee \rightarrow \text{Behind}(x, y) \vee \rightarrow \text{Green Signal}(y)$
 $\vee \text{Proceed}(x)$

Agricultural Expert Reasoning System:

P1: Soil Dry $\rightarrow$ Irrigation Needed

P2: Irrigation Needed $\wedge$ Crop wheat $\rightarrow$ Apply Drip Method

P3: $\rightarrow$ Apply Drip Method $\wedge$ Crop wheat $\rightarrow$ Crop at risk

P4

Soil Dry = True

Crop wheat = True

Convert to CNF

Soil Dry $\rightarrow$ Irrigation Needed

Eliminate implication:

$\rightarrow$ Soil Dry $\vee$ Irrigation Needed

CNF:

$(\rightarrow \text{Soil Dry} \vee \text{Irrigation Needed})$

...implication.

Apply Drip Method $\vee \rightarrow$ Crop wheat $\vee$ Crop at risk

Resolution Refutation:

Relevant clauses:

$\rightarrow$ Soil Dry $\vee$ Irrigation Needed

$\rightarrow$ Irrigation Needed $\vee \rightarrow$ Crop wheat $\vee$ Apply Drip Method

Soil Dry

Crop wheat

$\rightarrow$ Apply Drip Method

Resolution:

From (1) and (3)

Irrigation needed

Now resolve

Apply Drip method with $\rightarrow$ Apply Drip method

$\therefore$ (empty clause)

$\rightarrow$ Apply Drip method $\wedge$ Crop wheat

$\therefore$ (Crop at risk does not follow)

Wumpus World Knowledge Agent:

Stench at $[1,2]$

Breeze at $[1,1]$

Glitter at $[2,2]$

Rules:

Stench $[1,2] \rightarrow$ Wumpus is adjacent to $[1,2]$

Breeze $[1,1] \rightarrow$ Pit is adjacent to $[1,1]$

Glitter $[x,y] \rightarrow$ Gold is at $[x,y]$

$\neg$ Wumpus $[x,y] \wedge \neg$ Pit $[x,y] \rightarrow$ Safe $[x,y]$

Safety of Path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

At $[1,1]$ there is a Breeze, indicating a possible adjacent Pit

At $[1,2]$ there is a Stench, indicating a possible adjacent Wumpus.

There, the KB does not establish that $[1,1]$ and $[1,2]$ are safe

Gold is at $[2,2]$

Conclusion:

The agent should treat $[1,1], [1,2]$ as potentially dangerous and requires additional knowledge.